GFN1000 In Dialogue with Nature

与自然对话

JAMES D. WATSON

Common ancestor

- billions of years ago

Competition within species

- among individuals

Environmental selection

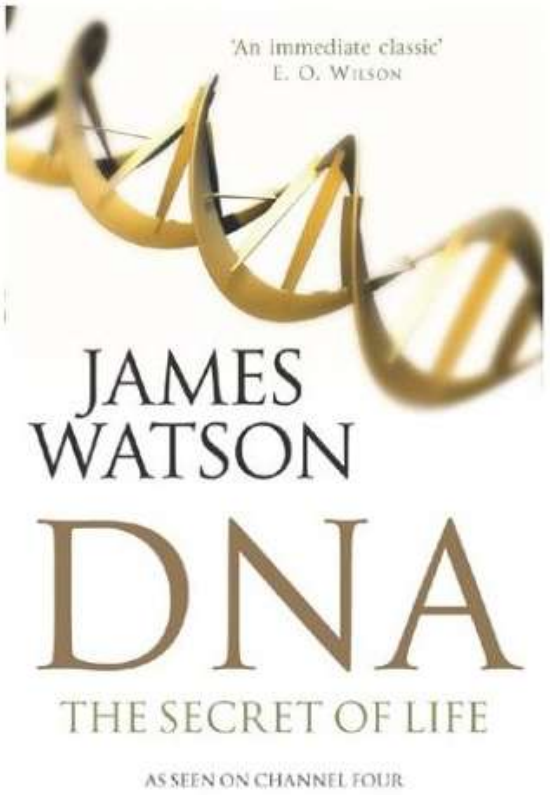
- survive or disappear

Heredity (Inheritance)

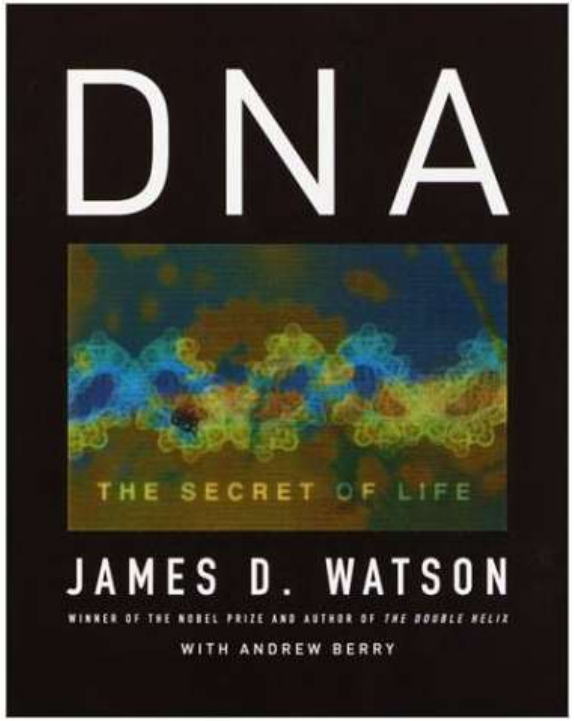
- parents to children

Core Question

What Is the Code of Life?



DNA: the Secret of Life



James D. Watson, *DNA: The Secret of Life*.

New York: Alfred A. Knopf, 2003.

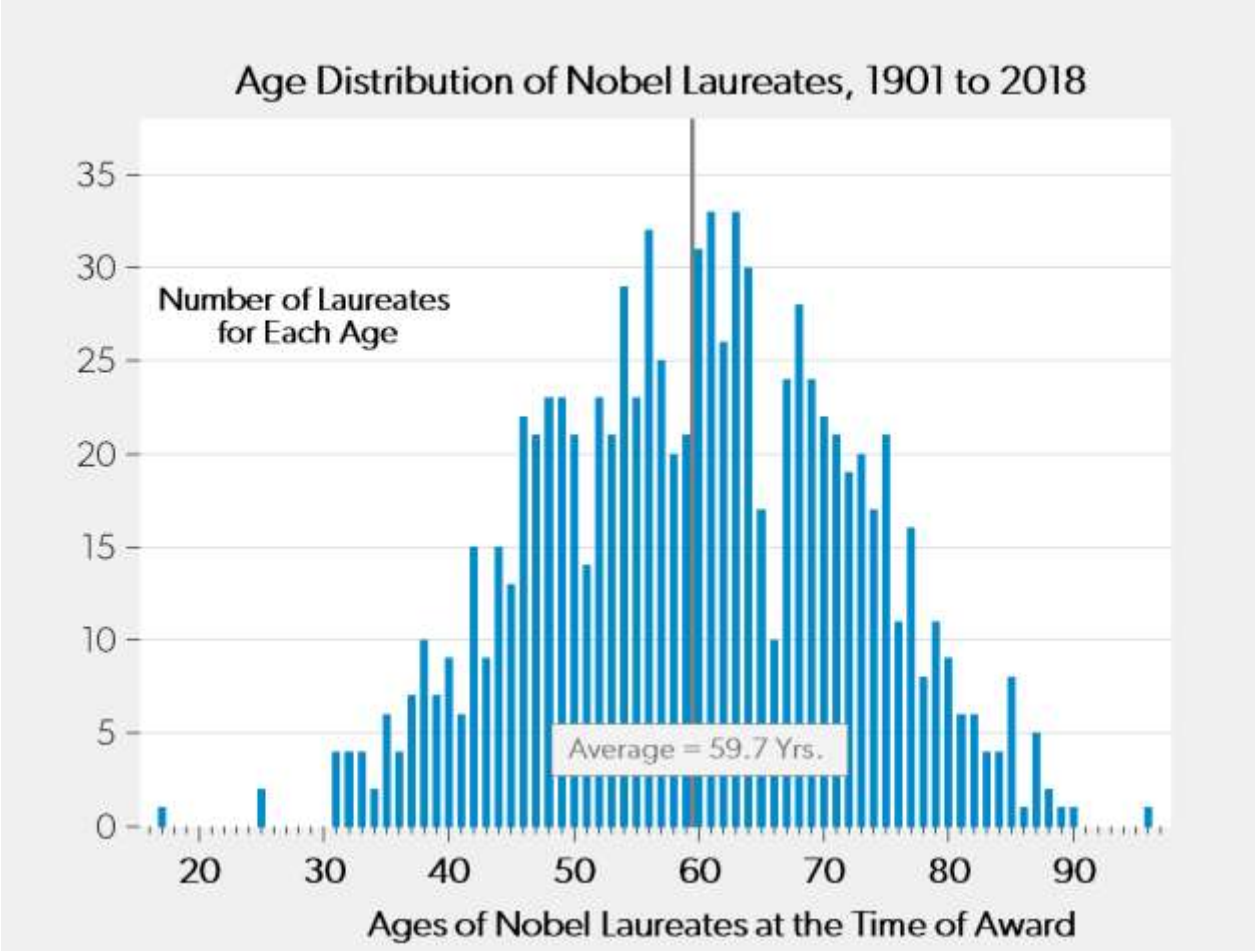
Read:

- Ch. 1. Beginnings of genetics: from Mendel to Hitler
- Ch. 2. The double helix: this is life

Question 1

What is the average age of the Nobel Laureates (1901-2018) when they were awarded the Nobel Prize?

- A. 39
- B. 49
- C. 59
- D. 69


Age Distribution of Nobel Laureates, 1901 to 2018
Number of Laureates for Each Age
35
30
25
20
15
10
5
0
Average = 59.7 Yrs.
20
30
40
50
60
70
80
90
Ages of Nobel Laureates at the Time of Award

Question 2

On average, Nobel Laureates of which category are the youngest?

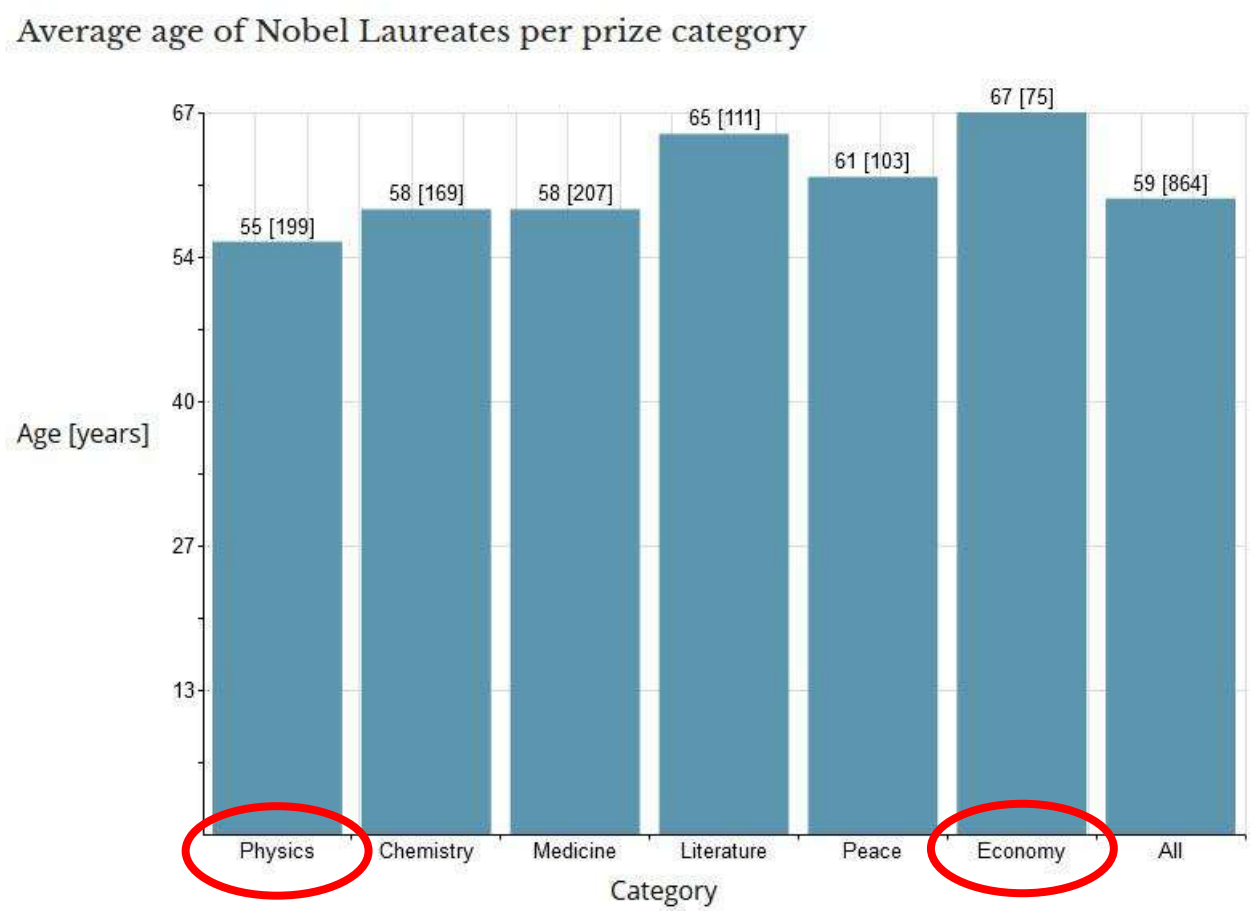
- A. Physics.
- B. Chemistry.
- C. Medicine.
- D. Literature.
- E. Peace.
- F. Economics.

Question 3

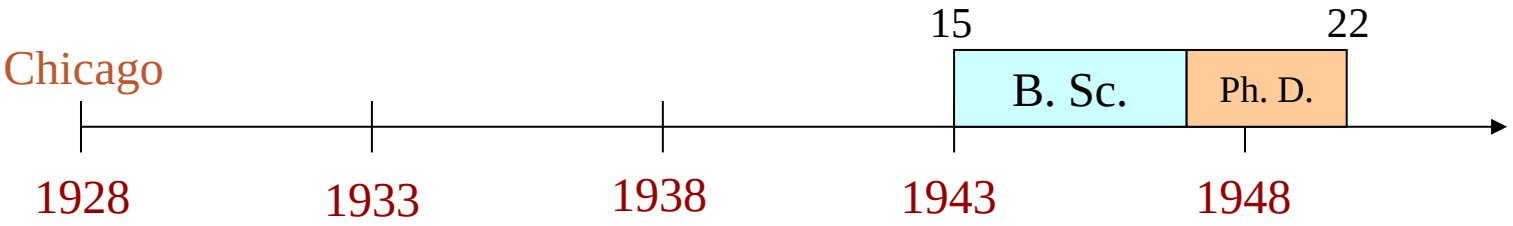
On average, Nobel Laureates of which category are the oldest?

- A. Physics.
- B. Chemistry.
- C. Medicine.
- D. Literature.
- E. Peace.
- F. Economics.

1901-2014


Average age of Nobel Laureates per prize category
Age [years]
67
54
40
27
13
55 [199]
58 [169]
58 [207]
65 [111]
61 [103]
67 [75]
59 [864]
Physics
Chemistry
Medicine
Literature
Peace
Economy
All
Category

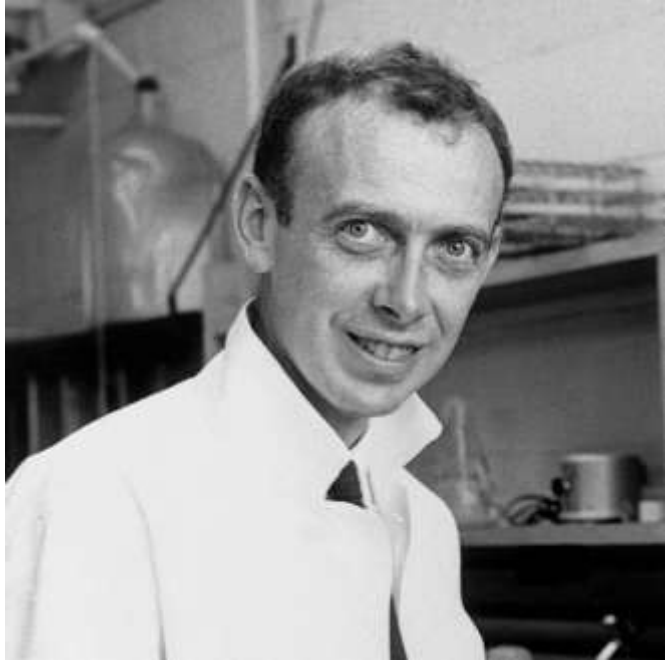
James D. Watson (1928-)





- Molecular biologist (分子生物学家)
- Entered U. of Chicago at 15 and got B. Sc. in Zoology at 19
- Got Ph. D. in Zoology at Indiana University in Bloomington at 22
- Became a Postdoc at Cambridge at 23
- Discovered double helix structure of DNA (with Crick) at 24
- Awarded Nobel Prize in Physiology or Medicine (1962) at 34 with Francis Crick and Maurice Wilkins, for their discoveries concerning the molecular structure of nucleic acids (核酸的分子结构) and its significance for information transfer in living material.
- Director of Cold Spring Harbor Laboratory at 40
  - CSHL is ranked among the leading basic research institutions in molecular biology and genetics.

James D. Watson



"The nations of the world must see that the human genome belongs to the world's people, as opposed to its nations."

He resigned as head of Human Genome Project at NIH after this (1992).

"If you are really stupid, I would call that a disease."

"[I am] inherently gloomy about the prospect of Africa [because] all our social policies are based on the fact that their intelligence is the same as ours—whereas all the testing says not really."

He was forced to resign as Chancellor of Cold Spring Harbor Laboratory afterwards (2007).

"the most unpleasant human being I had ever met."
– E. O. Wilson, Harvard University

"[*DNA: The Secret of Life*] an immediate classic."
– E. O. Wilson

Chapter 1

Beginnings of Genetics: From Mendel to Hitler.

- Para. 1-10: Introduction.
- Para. 11-29: Mendel and his influence.
- Para. 30-42: Social consequence --- Hitler.

- Francis Galton

Long history of genetics

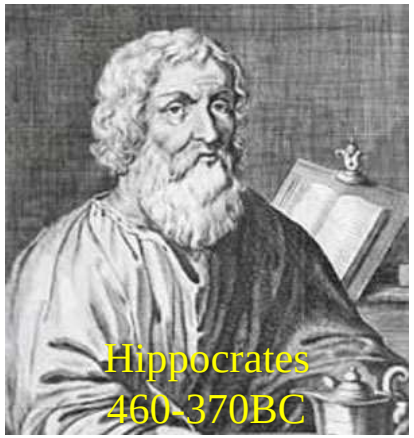
- Heredity – universal, observable
- Inherited attributes (body characters…)
- Genetic diseases (intermarrying within clan)
- Applied in improving domesticated animals (faster horse) and plants (bigger fruit).
- "Genetics" coined by William Bateson in 1909.

Why did it take so long to understand genetics?

- The field was not well developed:
  - Complicated [child ≠ (father + mother)/2].
  - Confused: does the feature come from heredity or development?

how a cross between a brown and black dog turns out, and all parents consciously or subconsciously track the appearance of their own characteristics in their children. One simple reason is that genetic mechanisms turn out to be complicated. Mendel's solution to the problem is not intuitively obvious: children are not, after all, simply a *blend* of their parents' characteristics. Perhaps most important was the failure by early biologists to distinguish between two fundamentally different processes, heredity and development. Today we understand that a fertilized egg contains the genetic information, contributed by both parents, that determines whether someone will be afflicted with, say, porphyria. That is heredity. The subsequent process, the *development* of a new individual from that humble starting point of a single cell, the fertilized egg, involves implementing that information. Broken down in terms of academic disciplines, genetics focuses on the information and developmental biology focuses on the use of that information. Lumping heredity and development together into a single phenomenon, early scientists never asked the questions that might have steered them toward the secret of heredity. Nevertheless, the effort had been under way in some form since the dawn of Western history.

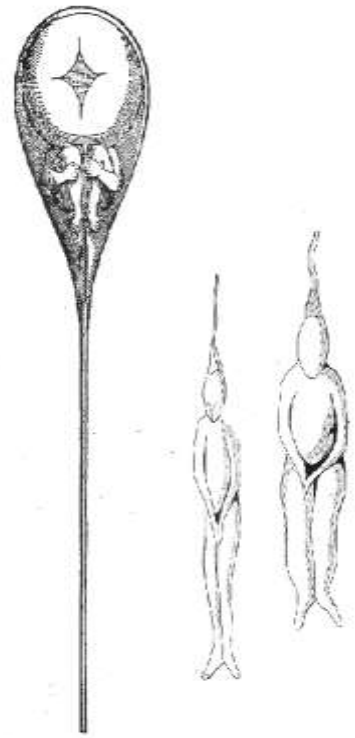
Before the discovery of DNA, heredity was explained as follows…

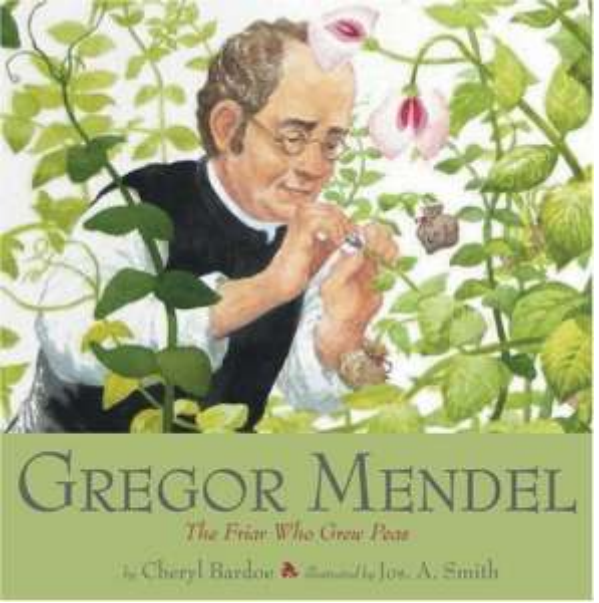

Hippocrates
460-370BC

- 'Father of Medicine'
- Pangenesis (泛生说): heredity = transfer of miniature body parts, embryos were assembled from a set of minuscule components (1, par. 8)
- Adopted by Lamarck (inheritance of acquired traits) and Darwin *etc.*
- Defeated by August Weismann's mice experiment (cutting tail)


Antonie van Leeuwenhoek
1632-1723

- "father of microbiology", the 1st microbiologist
- 1st to observe sperms with a microscope (1677)
- Preformationism (先成说) , which means that a preformed miniature person imagined to exist in the egg or the head of a sperm cell
- Early in 19th century, better microscopes defeated preformationism.

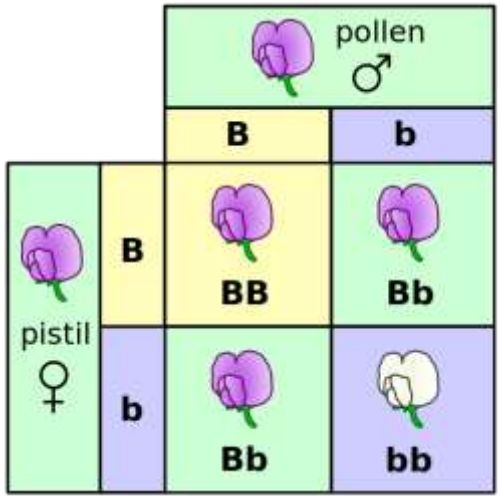

homunculus



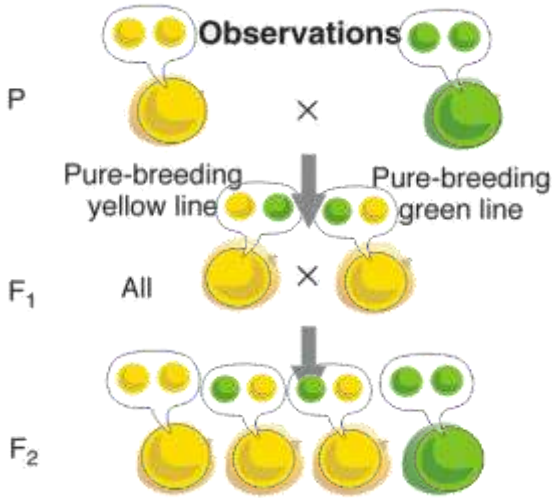
Gregor Mendel's work

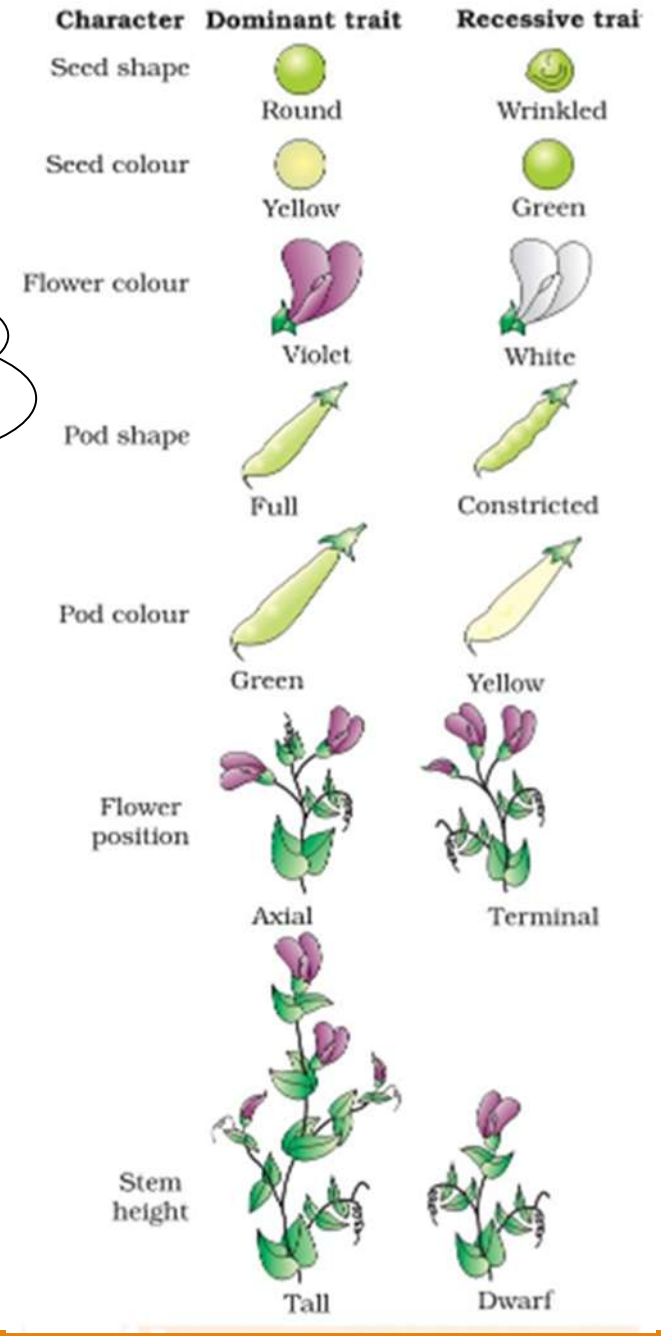
1856~1865: I crossbreed the plants, and did statistic analysis. → Quantitative study

(1822 – 1884)

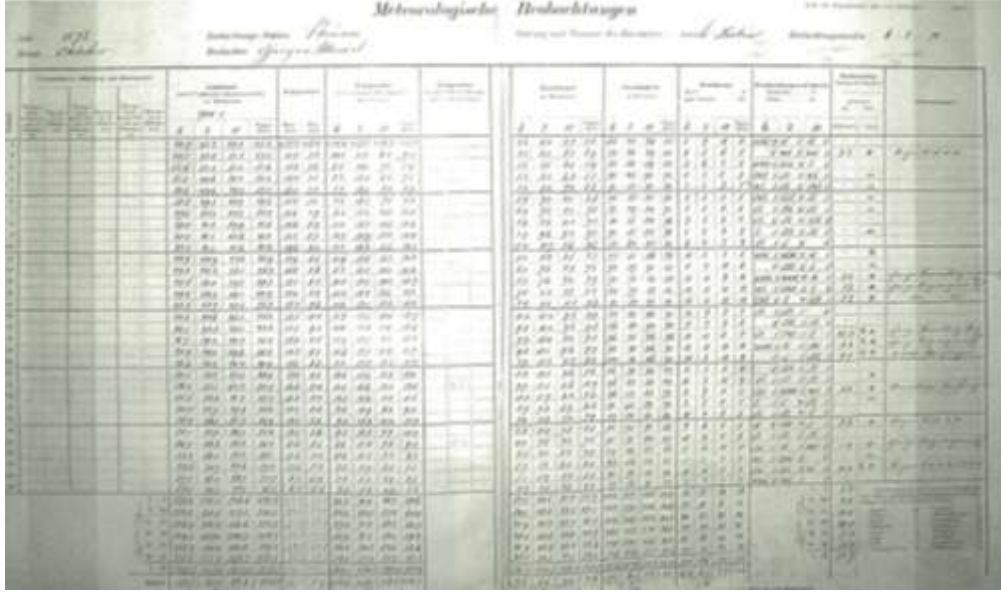


Genetics diagram: Punnett square describing one of Mendel's crosses, between parents that are heterozygous for the purple/white color alleles.





Gregor Mendel's work



Process:
1856-1865

Discovery:

15 Mendel realized that there are specific factors—later to be called "genes"—that are passed from parent to offspring. He worked out that these factors come in pairs and that the offspring receives one from each parent.

Lucky?
Use peas as the model plant.

Unlucky?
-1868: monk-teacher-researcher
1868-: abbot (男修道院院长) of the monastery (修道院)
1884: Mendel succumbed (死于) to a combination of heart and kidney disease.
1900: rediscovery of Mendel's work by three plant geneticists.

Question 1:
Where are the Specific Factors?

Where are the factors?

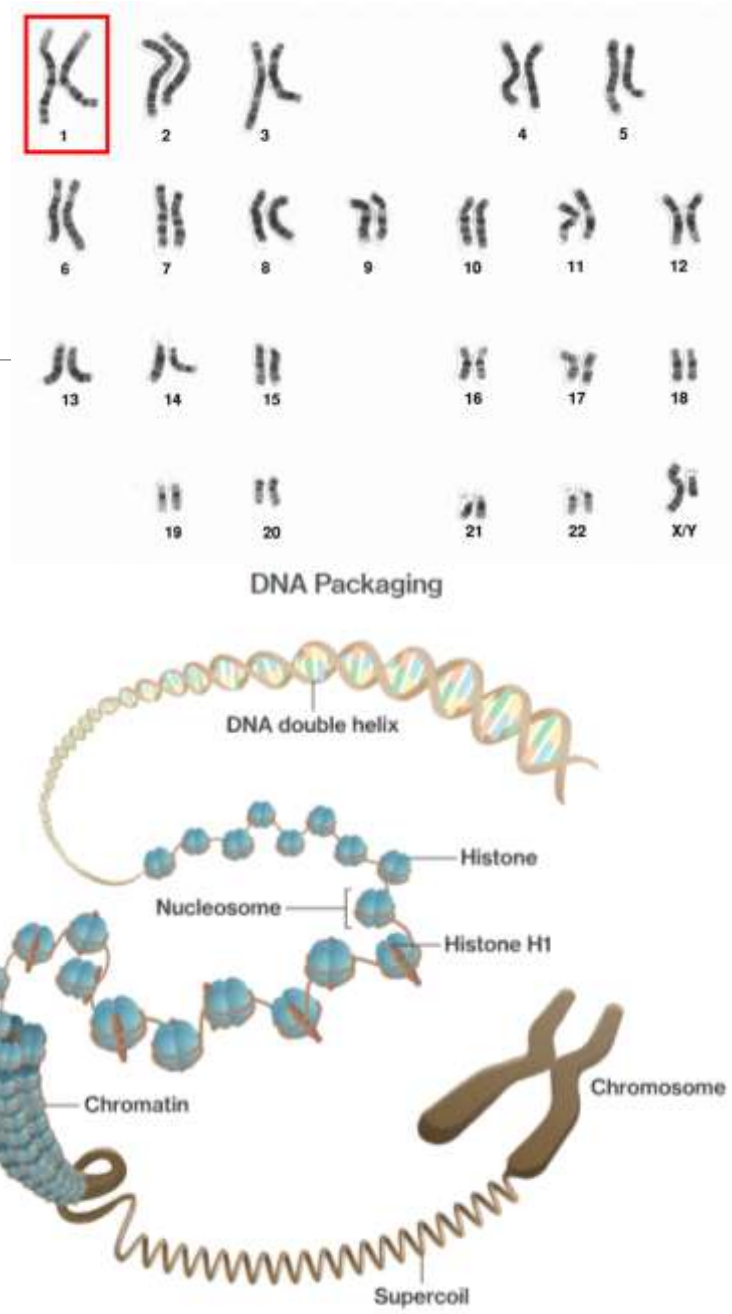
~ 1884: Discovery of **Chromosome** by ever-improving optics

– long stringy bodies in the cell nucleus

Chromosome

- Chromosome (染色体) is made up of a long stretch of DNA and proteins.

- Chromosome 1
  - The largest human chromosome.
  - Composed of 247 million DNA base pairs
  - 890 known diseases related to this chromosome, such as deafness, Alzheimer disease, glaucoma and breast cancer, *etc*.



Where are the factors?

~ 1884: Discovery of **Chromosome** – long stringy bodies in the cell nucleus by ever-improving optics

~ 1902: Sutton-Boveri's chromosome theory of inheritance

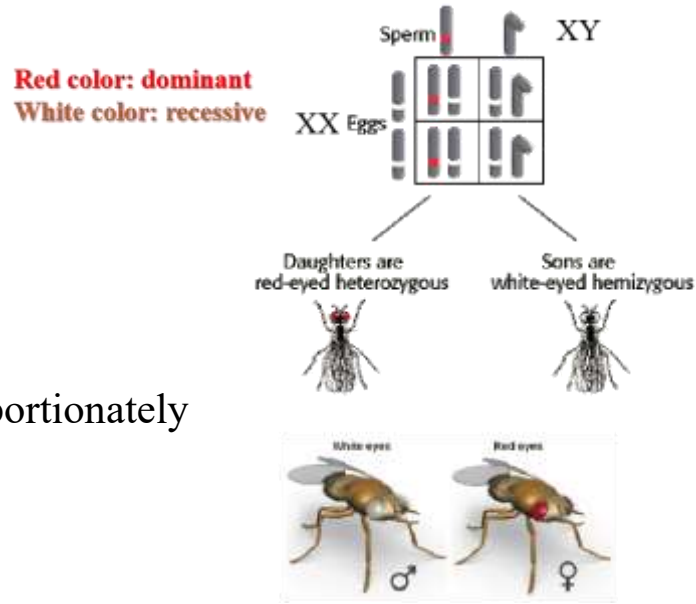
Related chromosome to the "specific factors" proposed by Mendel

~ 1915: T. H. Morgan

- Skeptical first
- Experiments with fruit fly (eye colors)
- Found one example of "sex-linkage", in which particular characteristic disproportionately represented in one sex.
- Proved the Sutton-Boveri theory



Thomas Hunt Morgan
1866-1945



How to explain variation in nature?

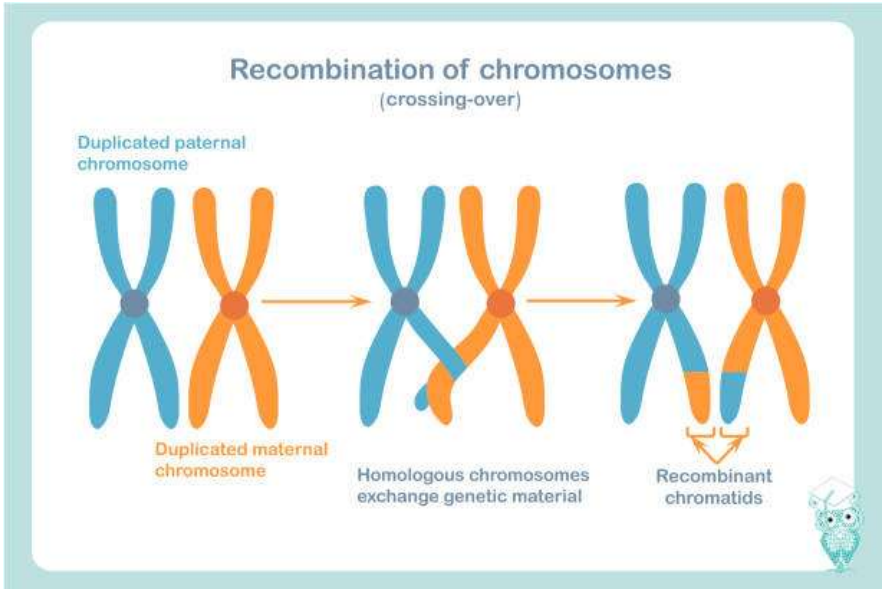
Sex linkage + Recombination (~1915) (I, par.23-29)



Thomas Hunt Morgan
1866-1945

1. **Prove Sutton-Boveri theory**
2. **Discover sex-linkage**
3. **Recombination重組**

Recombination: Chromosomes break apart and re-form during production of sperms and egg cells



4. Gene mapping

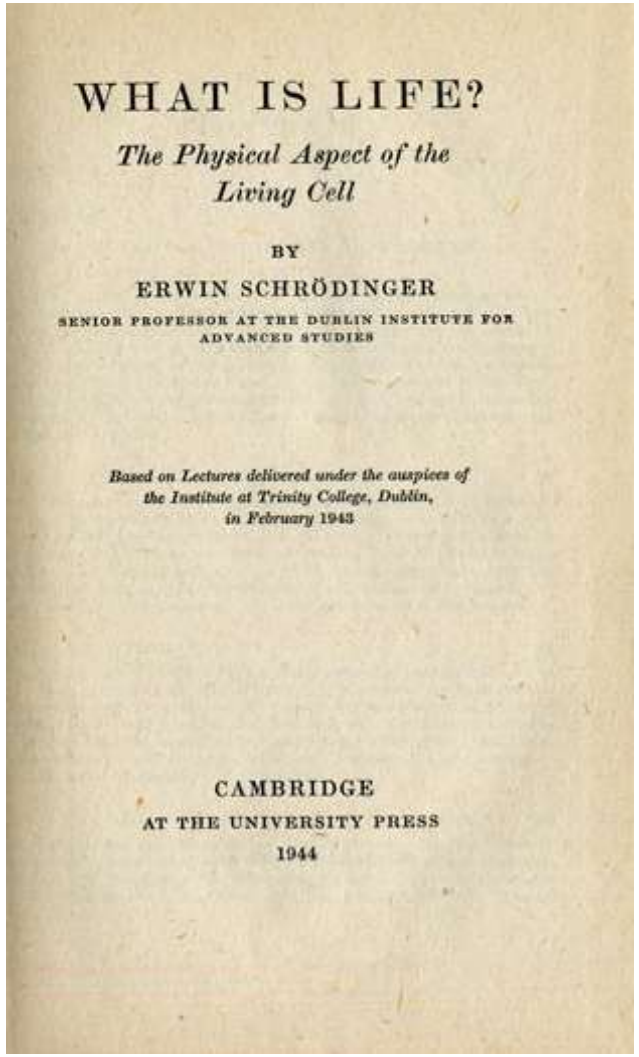
Determining the order and relative distance between genetic markers on a chromosome based on their pattern of inheritance.

CHAPTER 2

The double helix: this is life

CHAPTER TWO
THE DOUBLE HELIX: THIS IS LIFE

I got hooked on the gene during my third year at the University of Chicago. Until 1
then, I had planned to be a naturalist and looked forward to a career far removed from the urban bustle of Chicago's South Side, where I grew up. My change of heart was inspired not by an unforgettable teacher but a little book that appeared in 1944, *What Is Life?*, by the Austrian-born father of wave mechanics, Erwin Schrödinger. It grew out of several lectures he had given the year before at the Institute for Advanced Study in Dublin. That a great physicist had taken the time to write about biology caught my fancy. In those days, like most people, I considered chemistry and physics to be the "real" sciences, and theoretical physicists were science's top dogs.


WHAT IS LIFE?

The Physical Aspect of the Living Cell

BY

ERWIN SCHRÖDINGER

SENIOR PROFESSOR AT THE DUBLIN INSTITUTE FOR ADVANCED STUDIES

Based on Lectures delivered under the auspices of the Institute at Trinity College, Dublin, in February 1943

CAMBRIDGE

AT THE UNIVERSITY PRESS

1944

What is Life?

- Austrian theoretical physicist
- Nobel Prize in Physics, 1933, “for the discovery of new productive forms of atomic theory” (quantum mechanics) 量子力学
- *What is Life?* (1944)
- Read by Watson, Crick and Wilkins (Nobel Prize)!!!



Erwin Schrödinger

(1887-1961)

Influence of Erwin Schrödinger's *"What is Life"*?

Schrödinger argued that **life could be thought of in terms of storing and passing on biological information**. .. To understand life, then, we would have to identify these molecules, and crack their code. …

Paragraph 2, Chapter 2, *DNA*

In my own case, Schrodinger struck as chord because I too was intrigued by the essence of life. … the notion that life might be perpetuated by means of an instruction book inscribed in a secret code appealed to me. … And what sort of molecular trick could ensure that the code is exactly copied every time a chromosome duplicates?

Paragraph 3, Chapter 2, *DNA*

Question 1:
Where are the Specific Factors?

Question 2:

But in what language is the information written?

DNA or Protein?



Exclusively on chromosome, 4 subunits

20 different building blocks

1869 **DNA first isolated**

1919

1928 **Heat-killed S + Living R = Living S**

1937

1944

1947

1950

1952

Early 1953

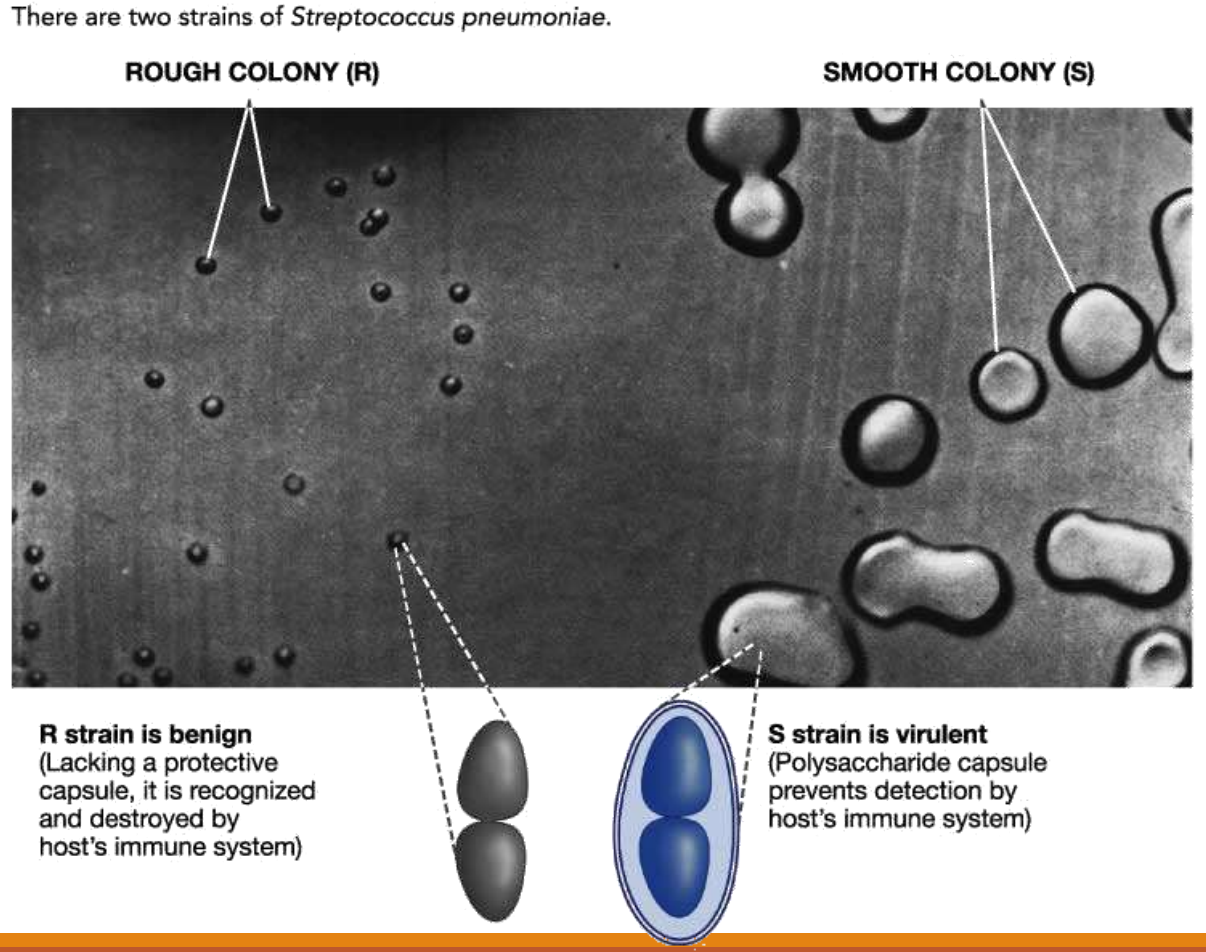
1953

1953

1958

Fred Griffith's experiment (1928)

Pneumonia (肺炎): he worked with 2 strains of the bacteria *Streptococcus pneumoniae* (*Pneumococcus*) 肺炎球菌



Rough: Harmless Smooth: Harmful

1869 DNA first isolated

1919

1928 Heat-killed S + Living R = Living S

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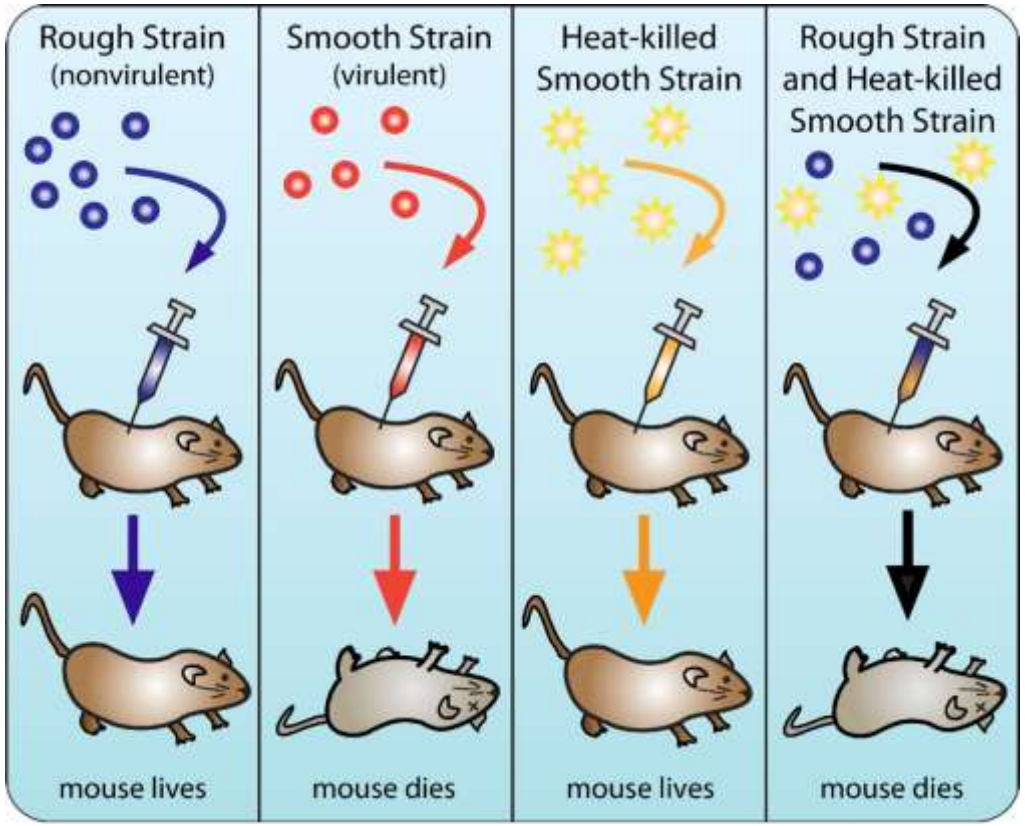
1958

Fred Griffith's experiment (1928)

He worked with 2 strains of the bacteria *Pneumococcus* 肺炎球菌

R: harmless

S: harmful



Heat-killed S + Living R → Living S

Transformation

Conclusions:

8 Through his involvement with public health, Griffith knew that multiple strains had sometimes been isolated from a single patient, and so he was curious about how different strains might interact in his unfortunate mice. With one combination, he made a remarkable discovery: when he injected heatkilled S bacteria (harmless) *and* normal R bacteria (also harmless), the mouse died. How could two harmless forms of bacteria conspire to become lethal? The clue came when he isolated the *Pneumococcus* bacteria retrieved from the dead mice and discovered living S bacteria. It appeared the living innocuous R bacteria had acquired something from the dead S variant; whatever it was, that something had allowed the R in the presence of the heat-killed S bacteria to transform itself into a living killer S strain. Griffith confirmed that this change was for real by culturing the S bacteria from the dead mouse over several generations: the bacteria bred true for the S type, just as any regular S strain would. A *genetic* change had indeed occurred to the R bacteria injected into the mouse.

1869 **DNA first isolated**

1919

1928 **Heat-killed S + Living R = Living S**

1937

1944 **DNA as transforming factor**

1947

1950

1952 **DNA confirmed as transforming factor**

Early 1953

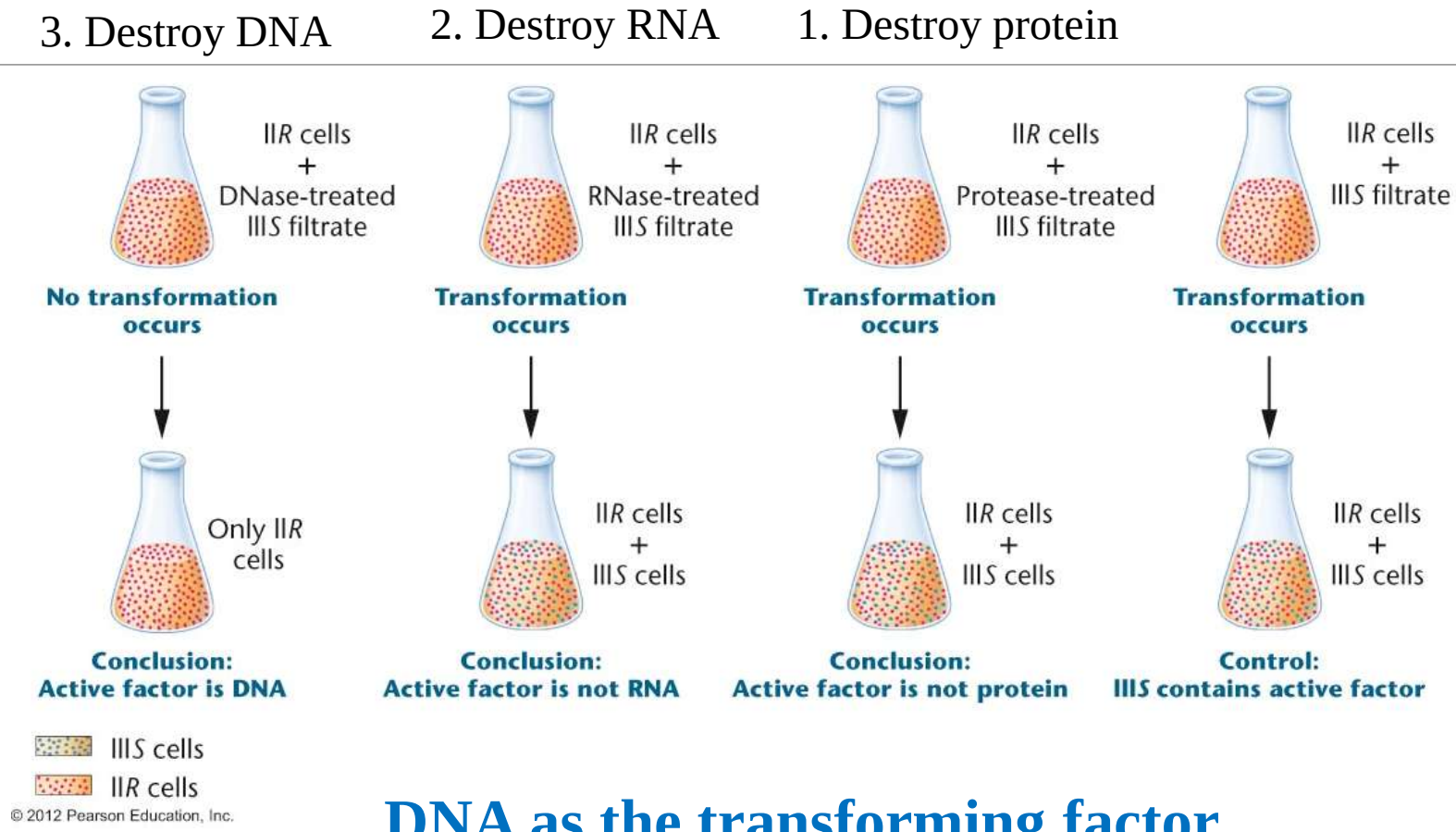
1953

1953

1958

Living harmless R + "destroyed" heat-killed harmful S ? Living S

Avery (埃弗里)-MacLeod-McCarty's experiment (1944)



DNA as the transforming factor

Later confirmed by Hershey and Chase (1952) *Para 41*

Question 1:
Where are the Specific Factors?

Question 2:

But in what language is the information written?

Question 3:

How do you describe the structure of DNA?

1869 DNA first isolated

1914-5 X Ray Diffraction

1928

1937

1944

1947

1950

1952

Early 1953

1953 DNA model & replication

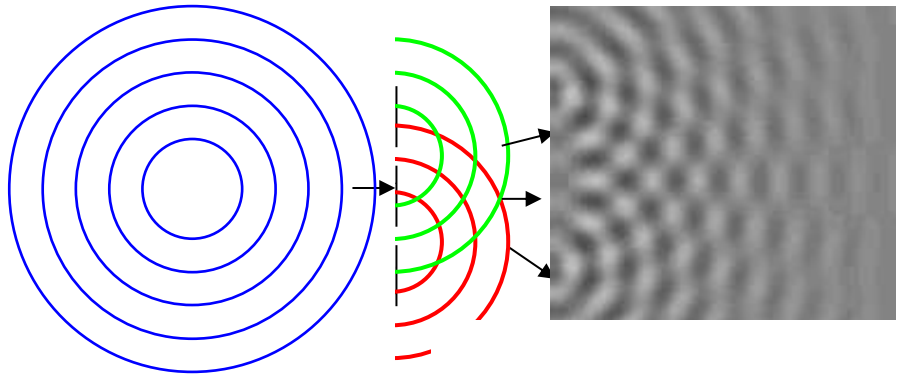
1953

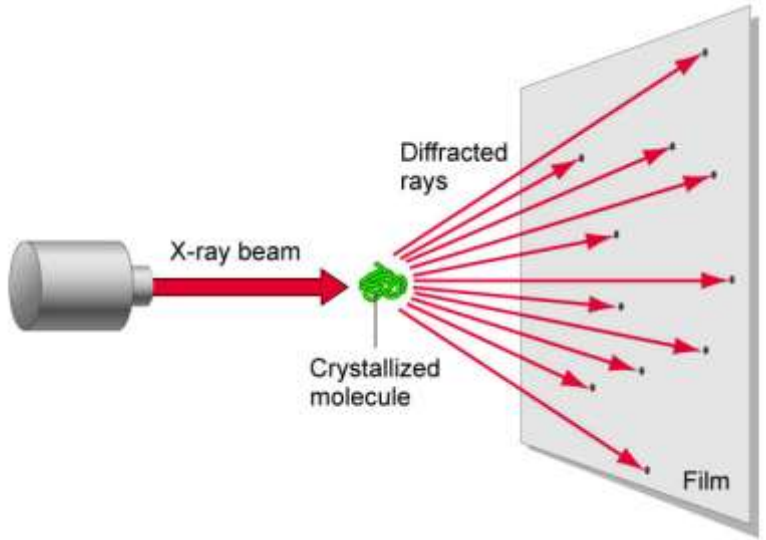
1958

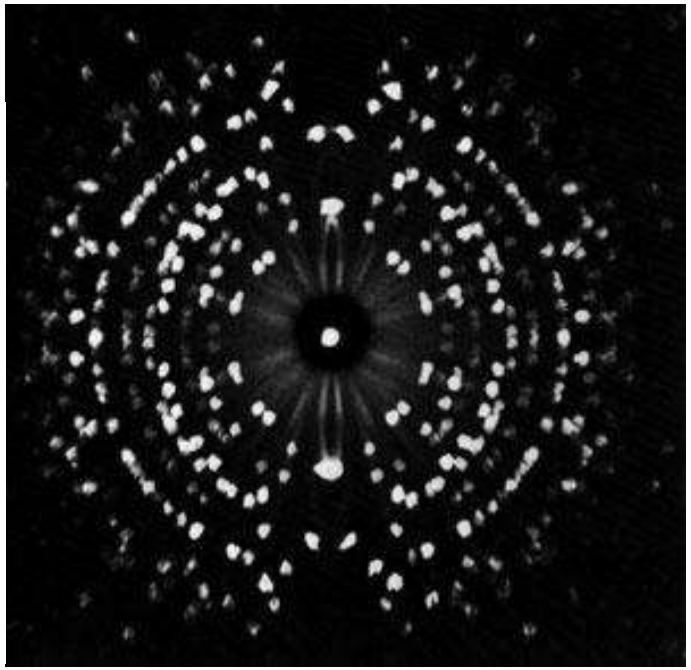
X-Ray Diffraction (XRD)

- Light has wave properties
- Interference of light

- X-ray: wavelength comparable to atomic distances in crystals
- Crystal structures have regular interference patterns
- Mathematics: **backward from interference patterns to crystal structures**

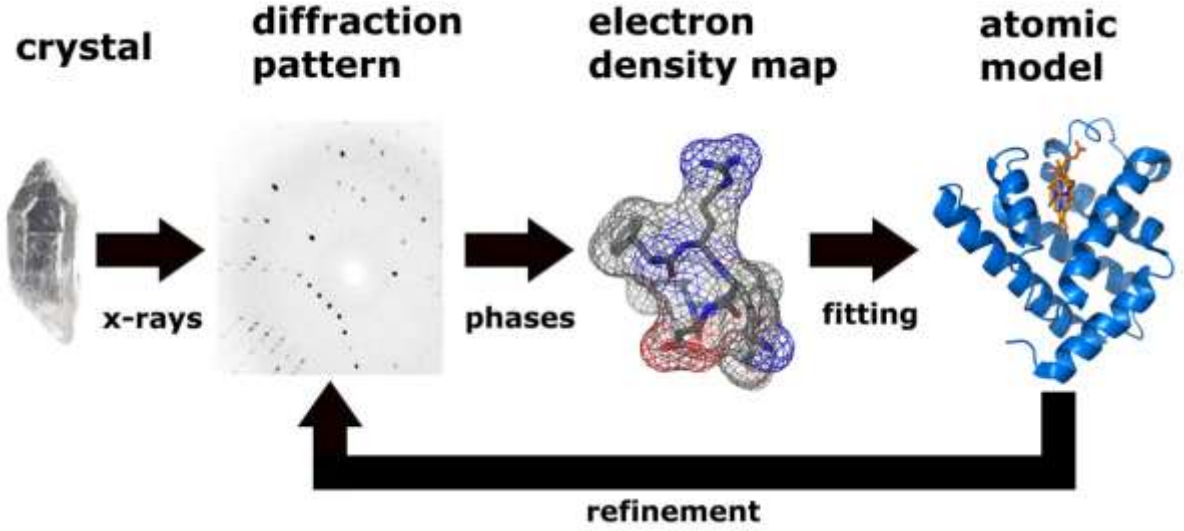






X-Ray Diffraction (XRD)





Workflow for solving the structure of a molecule by X-ray crystallography.

Discussions



Maurice Wilkins

All that remained to be settled is the number of chains!

Three-chain!

Backbone at the center!



Francis Crick

No!!! DNA is full of water!!! Backbone has to be on the outside!!!



Rosalind Franklin



James Watson

1869 DNA first isolated

1919 Components of DNA

1928

1937 DNA shown to be regular

1944

1947 Bases linked by hydrogen bond

1950 A=T G=C

1952

Early 1953

1953

1953

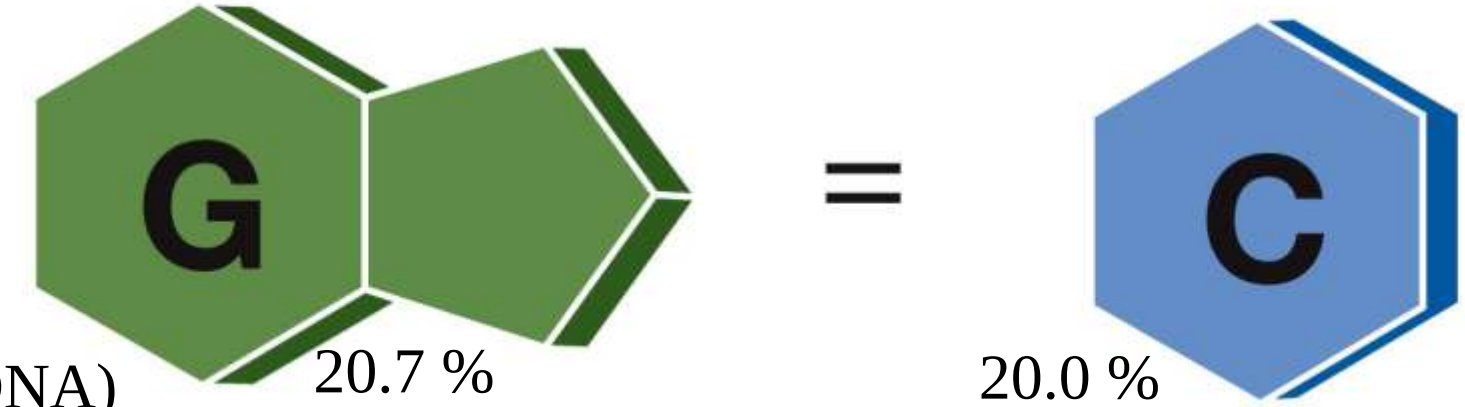
1958

Erwin Chargaff's rule (1950)



29.3 %

30.0 %

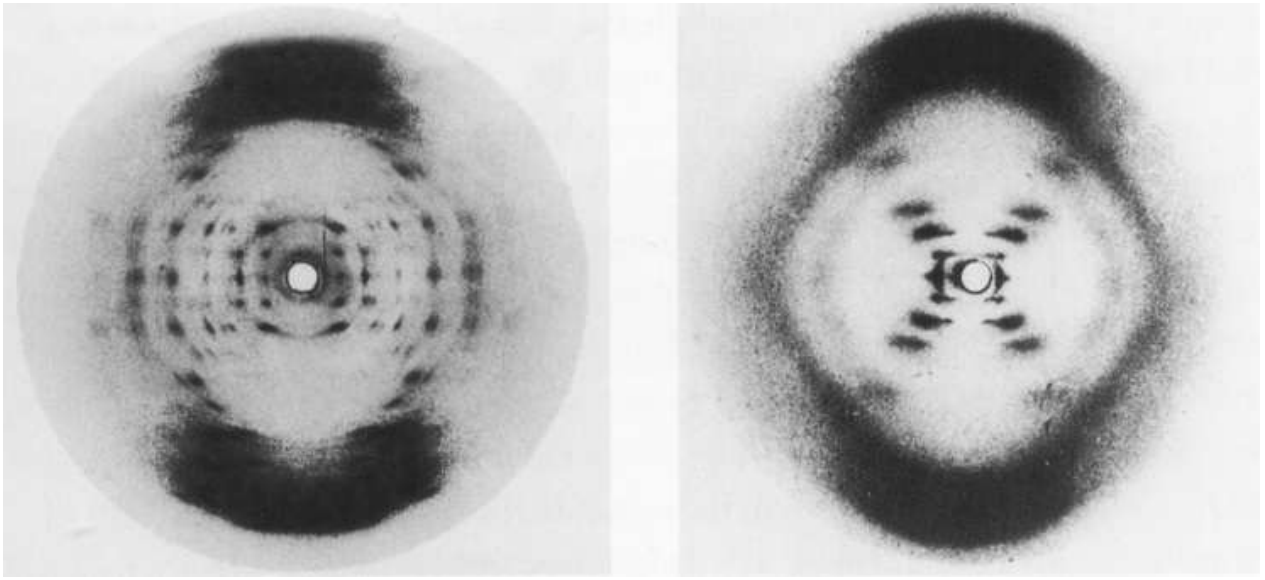


20.7 %

20.0 %

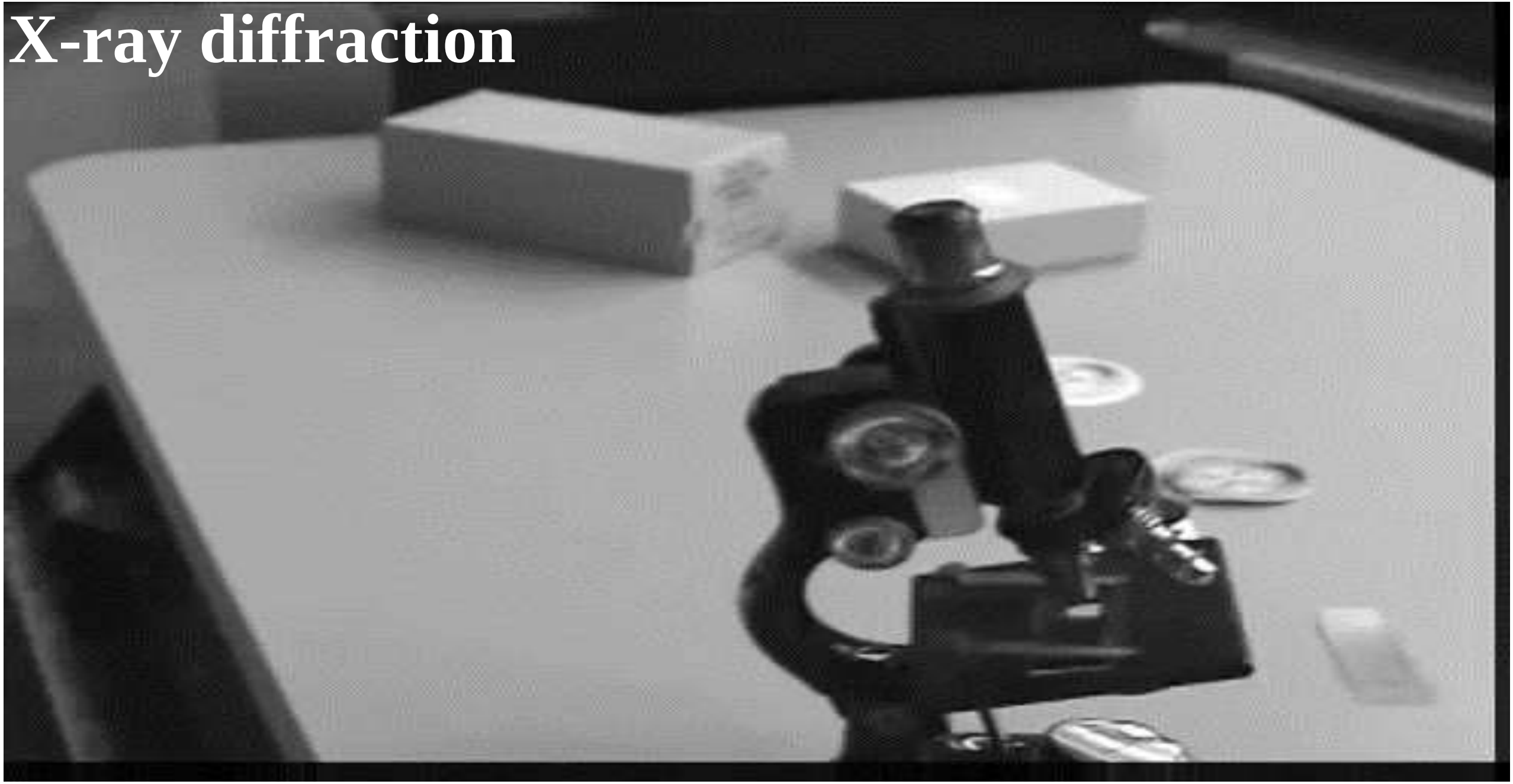
(Ratio of human DNA)

A and B forms of DNA (1953)



- X-ray photos of the A and B forms of DNA from, respectively, Wilkins and Franklin.
- The differences in molecular structure are caused by differences in the amount of water associated with each DNA molecule.

X-ray diffraction



Discussions



Maurice Wilkins

All that remained to be settled is the number of chains!

Three-chain!

Backbone at the center!



Francis Crick

No!!! DNA is full of water!!! Backbone has to be on the outside!!!



Rosalind Franklin

DNA density-measurement slightly favored a 2-chain model

As a biologist, I preferred the idea of a genetic molecule made of 2; after all, chromosomes are duplicating!



James Watson

A Jigsaw Puzzle of DNA

1869 — DNA first isolated

XRD 1914-5 1919 — Components of DNA

1928 — Heat-killed S + Living R = Living S

1937 — DNA shown to be regular

1944 — DNA as transforming factor

1947 — Bases linked by hydrogen bond

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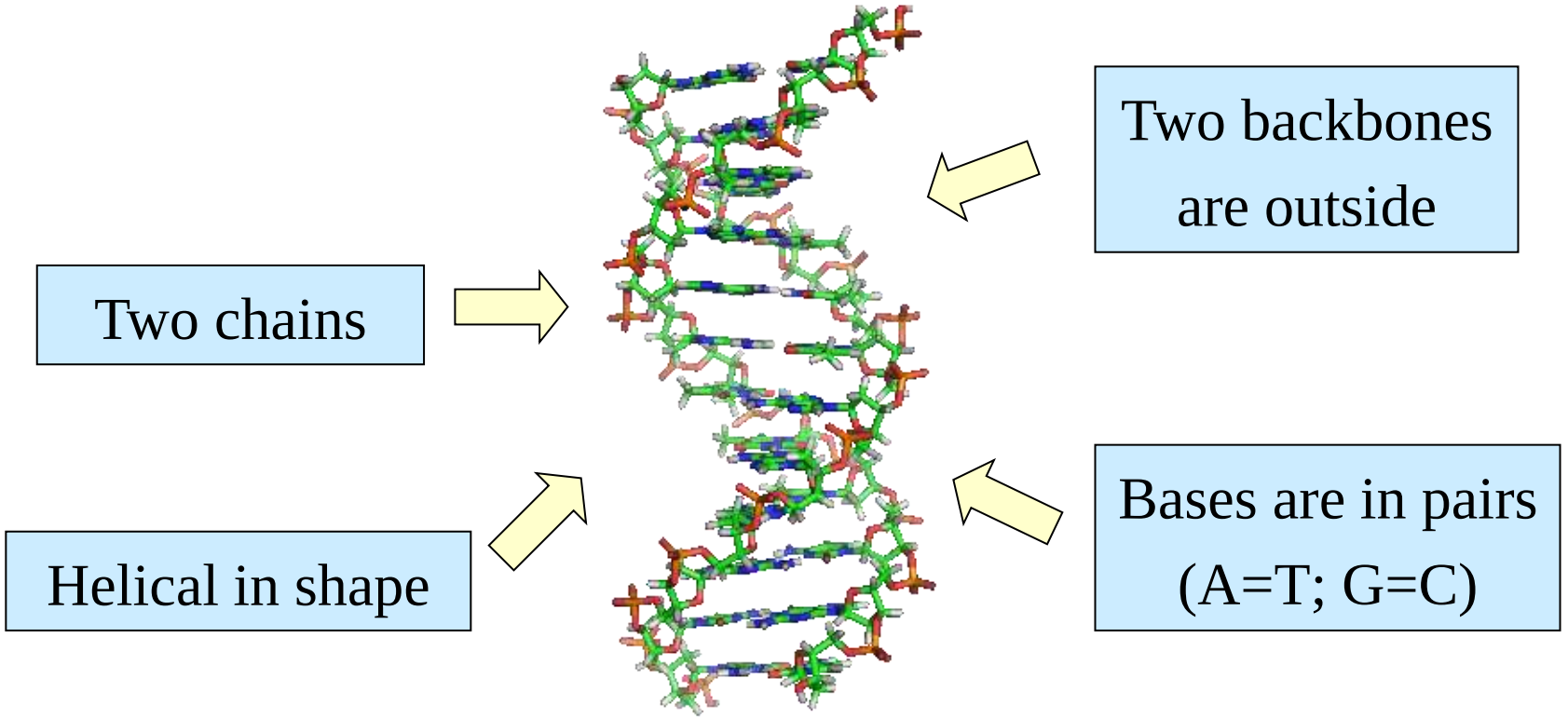
Early 1953 — Linus Pauling's paper

1953 — DNA model & replication

1953

1958

The two chains of the double helix structure is anti parallel, where they go in opposite directions but they bridge with these nitrogenous bases pairing with each other.





Francis Crick (1916–2004), English molecular biologist

Crick wrote a letter to his son, Michael on March 19, 1953

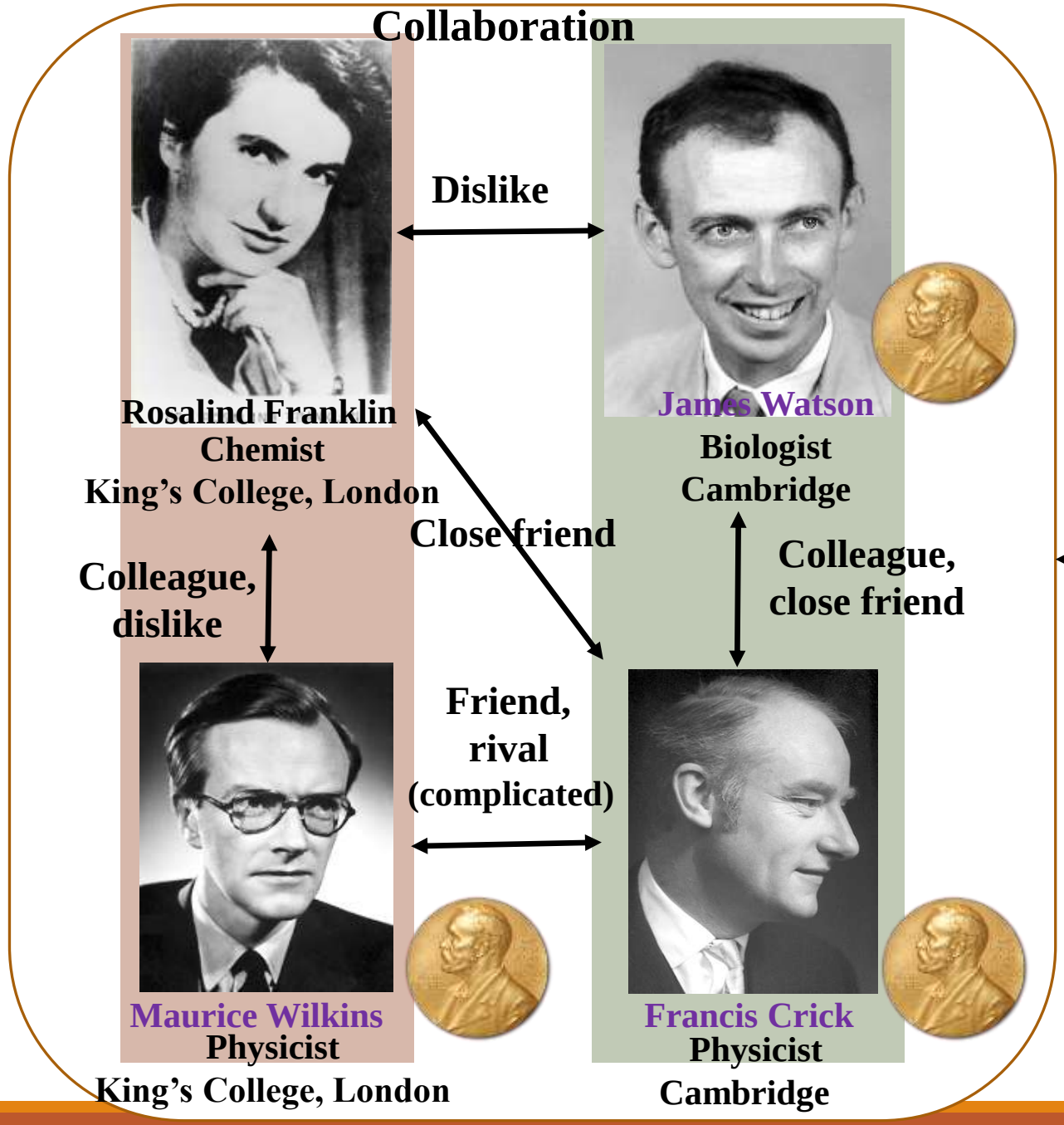
My Dear Michael, We've Discovered DNA

Writing to his son, Michael, Francis Crick tells of a "most important discovery."

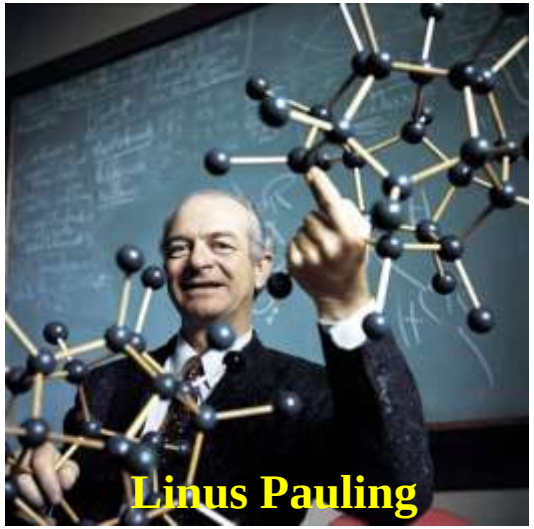
Son, Jim Watson and I have probably made a most important discovery.

2013, the letter was sold at a New York auction for a record price of 6 million dollars. The letter was purchased by an anonymous buyer who made the bid over the phone.

Competition and collaboration…



Rival



Chemist
Caltech, USA



<the nature of the chemical bond>

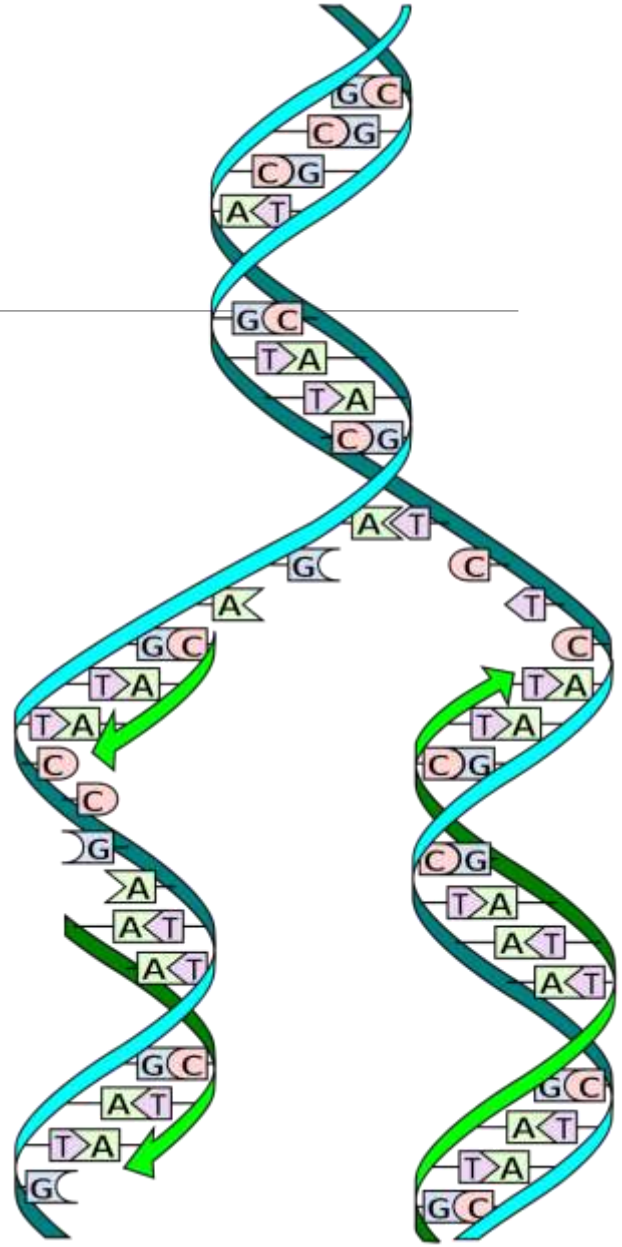
Next Question :

How does DNA pass the genetic information?

How does DNA replicate?

49 It was quite a moment. We felt sure that this was it. Anything that simple, that elegant just had to be right. What got us most excited was the complementarity of the base sequences along the two chains. If you knew the sequence—the order of bases—along one chain, you automatically knew the sequence along the other. It was immediately apparent that this must be how the genetic messages of genes are copied so exactly when chromosomes duplicate prior to cell division. The molecule would "unzip" to form two separate strands. Each separate strand then could serve as the template for the synthesis of a new strand, one double helix becoming two.

How to prove?



1869 DNA first isolated

1914-5 1919 Components of DNA

1928 Heat-killed S + Living R = Living S

1937 DNA shown to be regular

1944 DNA as transforming factor

1947 Bases linked by hydrogen bond

1950 A=T G=C

1952 DNA confirmed as transforming factor

Early 1953 Linus Pauling's paper

1953 DNA model & replication

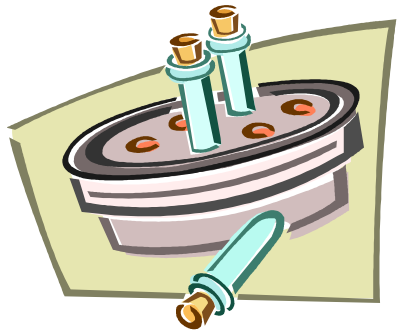
1953

1958

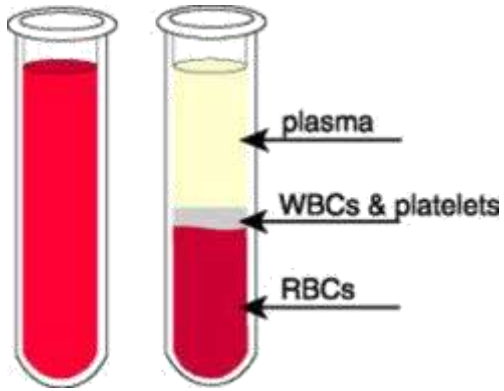
Meselson-Stahl experiment (1958) (梅瑟生-史达实验)

"**The most beautiful experiment in biology**"

- Operate by spinning at high speed (e.g. 14,000 times per min)
- Top layer: less dense (light); Bottom layer: denser (heavy)







- Nitrogen atoms (two forms: heavy ^{15}N and light ^{14}N) are a component of DNA.
- Bacteria were first grown in media with ^{15}N.
- Then a sample was taken and cultured in a media only contain ^{14}N.

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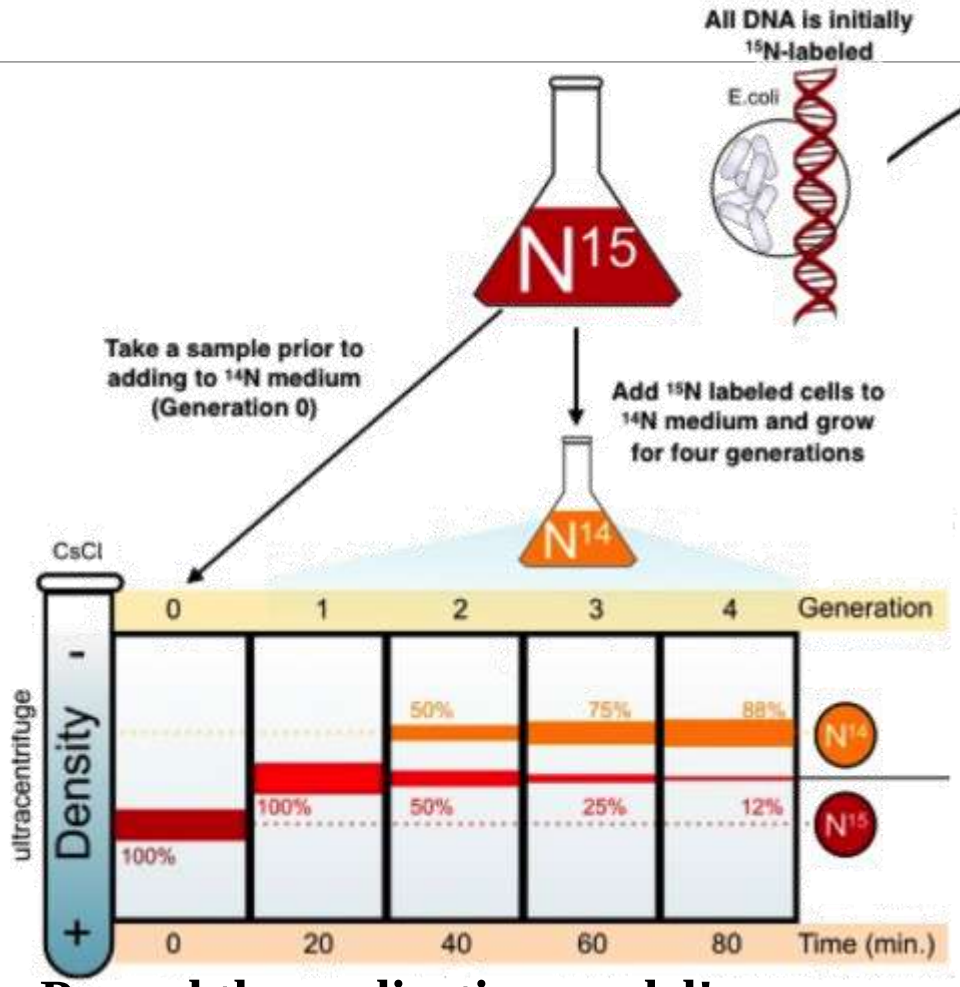
Early 1953 Linus Pauling's paper

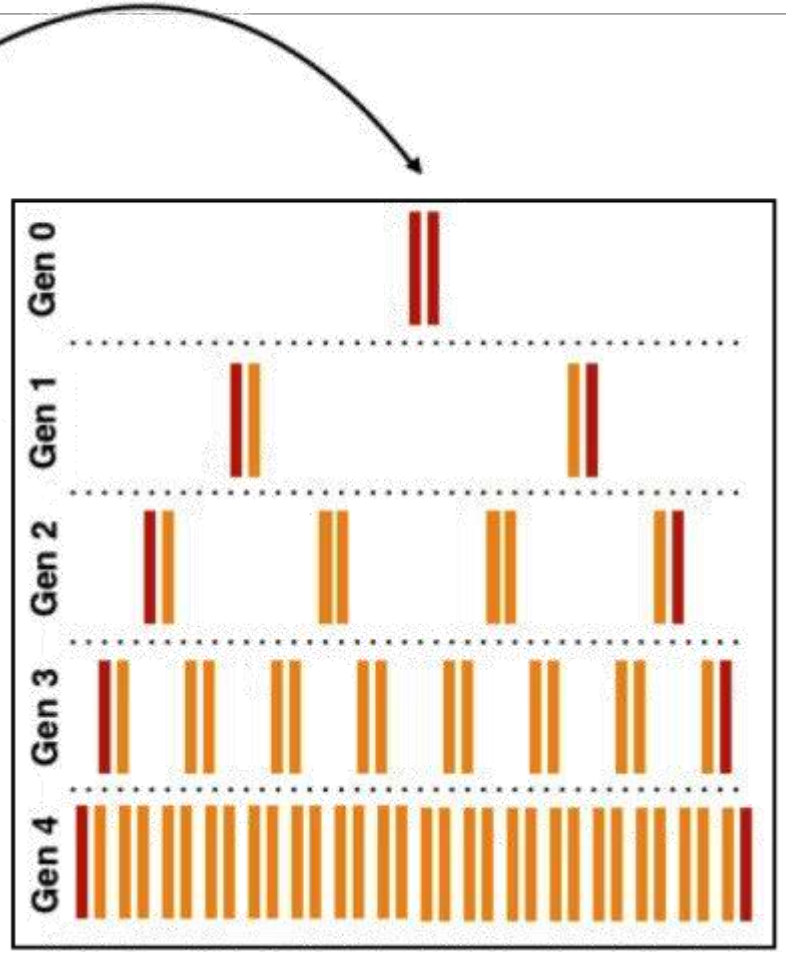
1953 DNA model & replication

1953

1958 Replication model proved

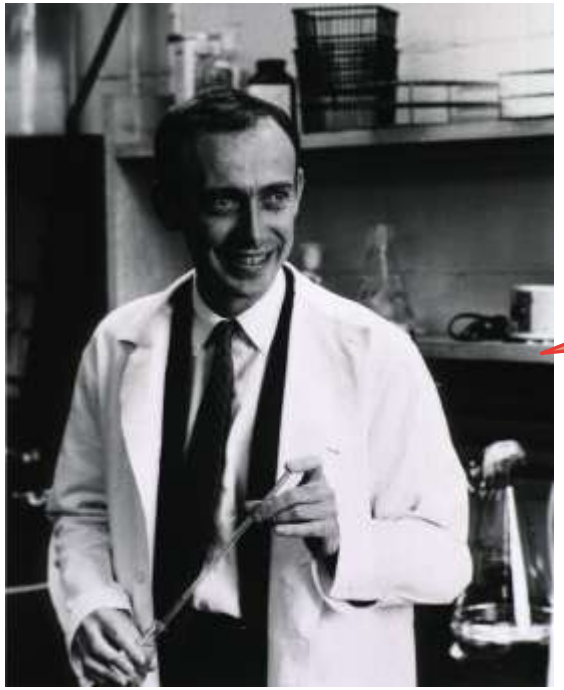
Meselson-Stahl experiment (1958) (梅瑟生-史达实验)





Proved the replication model!

Reading Assignment



Finish reading Text 5.
Quiz: texts of 3, 4 and 5.

Quiz in next tutorial !
Be punctual !